

TrES-3b

R-2057

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130404 · created 2026-08-09T00:50:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456386.8366 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.086 +/- 0.045
Transit depth (Rp/Rs)^2	0.74 +/- 0.78 [%]
Orbital Inclination (inc)	83.99 +/- 2.57
Airmass coefficient 1 (a1)	1.014 +/- 0.014
Airmass coefficient 2 (a2)	-0.0087 +/- 0.0046
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	#2 - [404, 49]
Optimal Aperture	3.62
Optimal Annulus	13.11
Transit Duration (day)	0.057 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.086 $\pm$ 0.045 vs 0.16309 (deviation 1.66 σ)
Duration fit vs published	0.057 d vs 0.0567 d (deviation 0.02 σ)
Mid-time O-C	1.6 min
Depth SNR	1.9
Residual scatter	1.36 %

Light curve

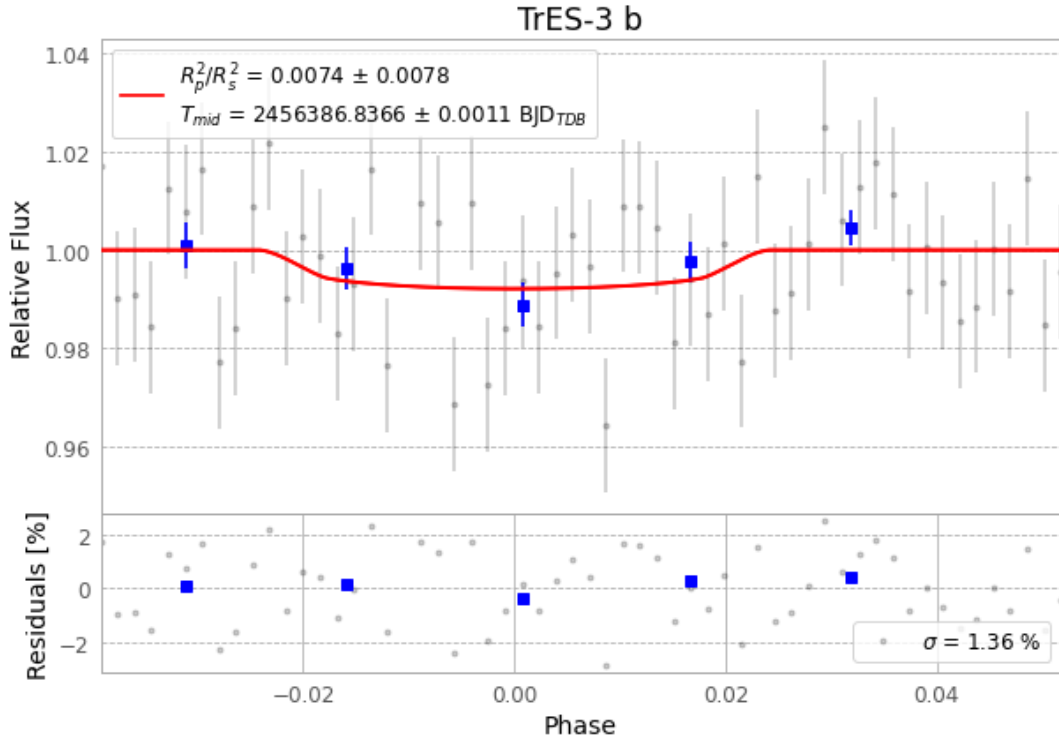


Figure 1 — FinalLightCurve_TrES-3 b_2013-04-03.png (SHA-256 c35f70ae77ae46ba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [199, 158], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4a0c45f0edfc919d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

58 FITS frames (38 MB), TRES-3130404064812.FITS → TRES-3130404093912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130404.log (dd679166283d...), FinalLightCurve_TrES-3 b_2013-04-03.png (c35f70ae77ae...), FinalLightCurve_TrES-3 b_2013-04-03.csv (606fa94c36f7...)

Compute resources

Wall time 308.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 00:50Z.